A Computational Framework for Comparative Analysis of Free Trade Agreements

A Policy-Facing NLP and LLM Workflow for Regional Trade Analysis

Diyouva Christa Novith

Executive framing: this project tests whether low-cost LLM workflows can reduce the time needed for policy analysts to compare complex free trade agreements, while remaining honest about where the workflow is useful, where it fails, and what still requires human legal review.

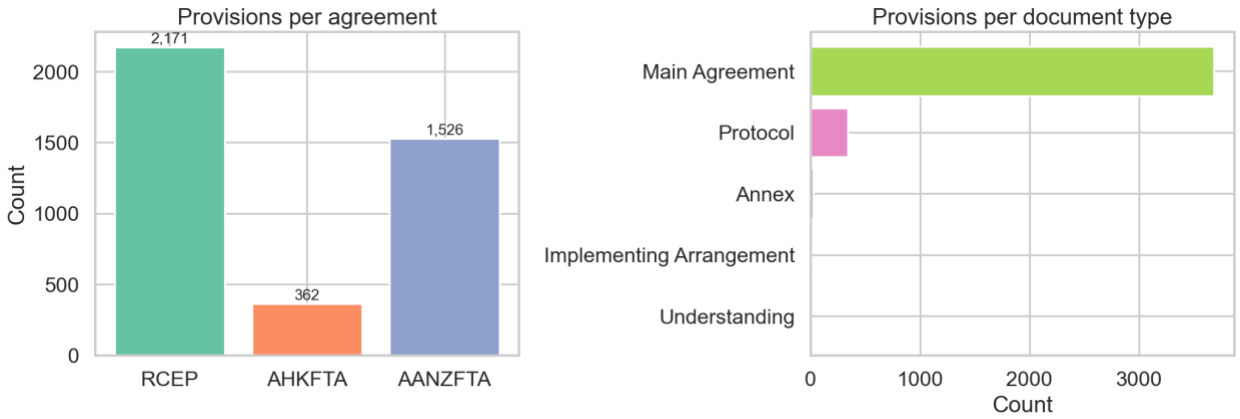

Figure 1. Corpus overview across the three Asia-Pacific FTAs.

Background

International trade in the Asia-Pacific is governed by a layered patchwork of bilateral and regional Free Trade Agreements. RCEP, AHKFTA, AANZFTA, CPTPP, and dozens of bilateral instruments overlap in coverage, with each agreement running to hundreds or thousands of legal provisions, distinct chapter structures, and quietly different drafting conventions. Trade economists describe this as the spaghetti bowl problem: the answer to a single practical question, such as whether a manufactured good qualifies for preferential tariff treatment under one agreement versus another, often turns on a 4-digit-versus-2-digit transformation rule buried inside an annex. For trade analysts, customs officers, and negotiators, comparing how two or three agreements treat the same topic is slow, manual work that does not scale with the rate at which the regional treaty network is expanding.

That bottleneck matters because the gap between what is documented and what is operational keeps widening. Exporters routinely operate under more than one preferential framework at the same time, compliance officers need to know where agreements diverge enough to disqualify the same shipment, and policymakers tracking regional integration need a structured view of where harmonisation is genuinely advancing rather than just rhetorically promised. Large Language Models open the possibility of automating the most labour-intensive layer of this work: extracting provisions from PDFs, classifying them into a shared policy taxonomy, and surfacing structural differences across agreements at scale. Whether the current generation of open-weight models is reliable enough to do this responsibly, where their accuracy holds and where it breaks, and how their outputs should be framed for downstream policy users are open empirical questions. This project investigates those questions on a corpus of 4,059 provisions extracted from three ASEAN-centred agreements, and positions the resulting pipeline as a first-pass triage layer for analysts rather than a substitute for legal review.

Project Scope and Research Questions

This project narrows the comparative-FTA problem to three ASEAN-centred agreements: RCEP, AHKFTA, and AANZFTA. Together they span the dominant contemporary trade architecture of Southeast Asia, and they differ enough in scope and drafting style to make a cross-agreement comparison meaningful: RCEP is the broadest fifteen-party instrument with twenty chapters; AHKFTA is a goods-focused bilateral framework between ASEAN and Hong Kong; AANZFTA is the older comprehensive agreement that introduced services, investment, and an independent dispute settlement mechanism. Seven source PDFs are processed end to end, yielding a corpus of 4,059 provisions classified against an eleven-category WTO and UNCTAD policy taxonomy. Two open-weight Large Language Models (LLaMA 3.3 70B and Qwen 3 32B) are run under three prompt strategies (zero-shot, few-shot, and chain-of-thought), producing six independent classification runs against a 50-provision hand-labelled gold set.

Three research questions organise the work, one per layer of the pipeline. RQ1 asks whether LLMs can reliably classify FTA legal provisions, and how that reliability shifts across model and prompt strategy combinations. RQ2 asks how the three agreements differ in observable design features, including thresholds, governance structures, and chapter scope. RQ3 asks whether the agreements show structural convergence or fragmentation in their treatment of key trade policy topics. The expected outcome is not a high-confidence legal classifier but a reproducible diagnostic layer: a stratified provision distribution per agreement, validation metrics that quantify where the classification can and cannot be trusted, an inter-run agreement signal across model and prompt configurations, and a structured cross-agreement comparison surfaced through retrieval-augmented narratives. Read together, these outputs give an analyst a defensible starting point for closer manual review and an evidence-backed picture of where regional harmonisation is actually advancing.

Data and Unit of Analysis

The project addressed three questions: whether LLMs can classify FTA provisions into a usable policy taxonomy, how the three agreements differ in issue emphasis and legal design, and whether some issue areas show regional convergence while others remain fragmented. The three agreements were RCEP, AHKFTA, and AANZFTA, chosen because together they span the main contemporary Asia-Pacific trade architecture.

Agreement	Rows / provisions	Share	Notes
RCEP	2,171	53.5%	Main agreement dominates corpus size
AANZFTA	1,526	37.6%	Includes protocol and implementing texts
AHKFTA	362	8.9%	Goods-focused agreement; annex-heavy and partially scanned
Total	4,059	100%	Final extracted provision corpus

The unit of analysis was the smallest meaningful legal provision that could be extracted cleanly from PDF source documents: article text, paragraph text, or a structured annex fragment. Each row stores an ID, agreement, document type, chapter, article, paragraph index, provision text, and character count. That level of granularity made it possible to classify, retrieve, and compare provisions without reducing the problem to whole-document summaries.

Model Comparison and Validation Findings

The project compared six model-strategy combinations on a shared 50-row hand-labelled gold set: LLaMA 3.3 70B and Qwen 3 32B, each run under zero-shot, few-shot, and chain-of-thought (CoT) prompting. The fixed evaluation cohort, identical prompt scaffolding, and identical category taxonomy were all chosen so that the six runs would be comparable on a like-for-like basis, with prompt strategy and model architecture as the only experimental variables. The headline result is not that one model won decisively, but that each strategy trades off overall hit rate and category balance in genuinely different ways, and that the choice between them is therefore an applied judgement call rather than a universal one.

Run	n	Accuracy	Macro-F1
LLaMA zero-shot	50	0.480	0.431
Qwen CoT	50	0.460	0.442
Qwen zero-shot	50	0.380	0.424
Qwen few-shot	50	0.380	0.373
LLaMA few-shot	50	0.340	0.336
LLaMA CoT	50	0.320	0.327

Three observations stand out from the table. First, no configuration crossed 50 percent accuracy, which is what justifies framing the pipeline as a triage tool rather than a fully trusted classifier; the validation ceiling is the single most important constraint on how the rest of the project should be read. Second, the most accurate run and the best Macro-F1 run are not the same. LLaMA zero-shot gets the most labels correct overall, while Qwen CoT distributes performance more evenly across the eleven categories, including the rarer ones that suffer most under accuracy-only scoring. That split is a genuine trade-off a policy user would need to understand: if downstream value comes from getting common categories right, LLaMA zero-shot is the better choice, but if balanced coverage of rare categories like SPS or Intellectual Property matters, Qwen CoT is the more defensible pick. Third, prompt strategy moves the two models in opposite directions on chain-of-thought: CoT raises Qwen's Macro-F1 by 1.8 points relative to its zero-shot baseline, but lowers LLaMA's by 10.4 points. Few-shot prompting hurts both. This asymmetry only becomes visible once all six combinations are scored against the same gold set, and it has a direct operational implication: prompt strategy is a per-model decision and cannot be transferred from one architecture to another by analogy.

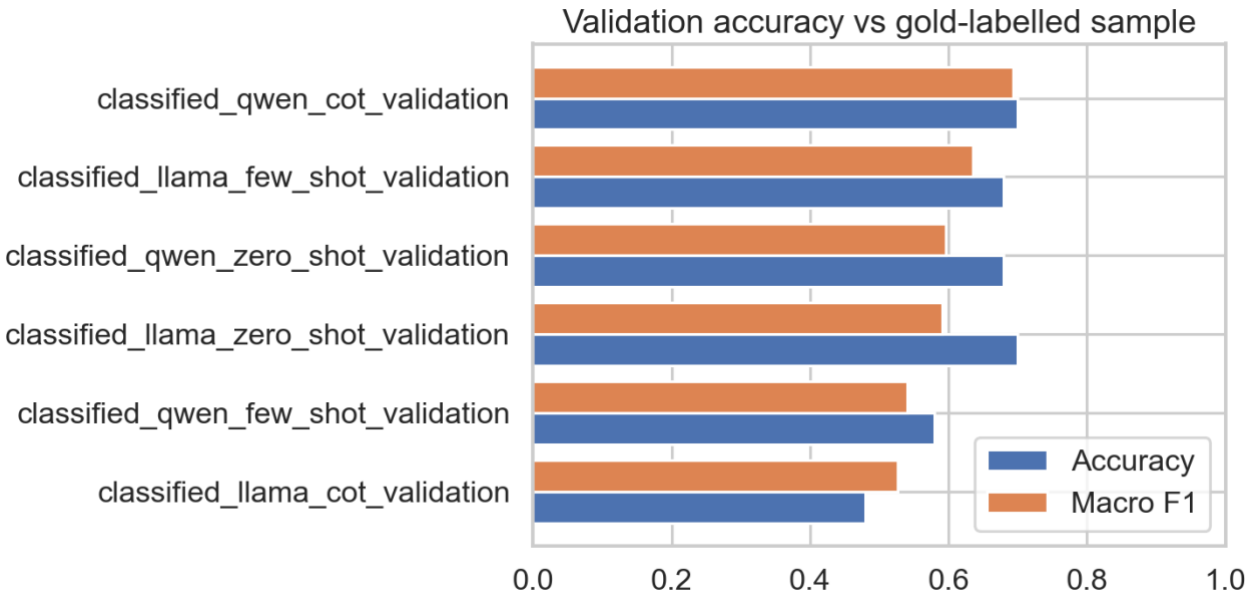

Figure 2. Validation performance across all six model-strategy combinations.

The repaired pairwise agreement analysis matters just as much as the validation table. Earlier drafts of the pipeline reported near-random Cohen's κ between some run pairs, which on first reading looked like model instability. Once the cohort-alignment bug was fixed and the κ was recomputed only on provisions where both runs labelled the exact same IDs,

agreement turned out to be moderate to substantial across the board. Cross-model zero-shot reaches κ = 0.702 on a shared cohort of 200 provisions; within-model zero-shot versus few-shot pairs reach 0.668 (LLaMA) and 0.689 (Qwen) on the same n; CoT-versus-CoT across the two models reaches 0.640 on n = 100; and the weakest pair, LLaMA few-shot versus Qwen few-shot, still sits at 0.582. None of these values are near random. Two implications follow. First, what looked like chaos between models was at least partly an evaluation-design problem rather than a model-behaviour problem; misaligned cohorts had inflated apparent disagreement, and once that artefact is removed the two models share a meaningful baseline signal. Second, that signal is more legible at zero-shot than at few-shot or CoT, which means prompt-engineering moves disrupt cross-model agreement rather than amplify it. The current bottleneck for trusting any single provision label is therefore not chaos between models, but low absolute validation quality against human labels: the six runs largely agree with each other, they just do not yet agree with the gold set strongly enough to support compliance-grade decisions. That is a methodologically clearer place to be than where the project started, and it sharpens the question for the next iteration from "are these models even doing the same task" to "how do we close the gap between consistent model behaviour and human-labelled ground truth."

Substantive Findings for a Policy Audience

Despite modest validation quality at the provision level, the project produced a set of policy-facing findings that hold up well at the aggregate level. That distinction matters: the system is more credible for pattern discovery across thousands of provisions than for any single provision label, and the findings below are framed accordingly. They describe shape and proportion of treaty content, where the signal is strong, rather than verdicts on specific clauses, where the signal is noisier. Read together, they give a policy audience a structured first pass at how three of the most consequential ASEAN-centred agreements compare on substance, scope, and procedural design.

- **AHKFTA is visibly goods-focused.** In the stratified cohort of 100 provisions per agreement, AHKFTA records zero classifications in Trade in Services, Dispute Settlement, and Intellectual Property. This is consistent with its design as a bilateral goods-trade instrument that delegates dispute resolution to the WTO DSU and that has no dedicated services, investment, or IP chapters at all. The absences are structural, not classification gaps.
- **AHKFTA is also markedly more rule-intensive on origin.** AHKFTA devotes 48 percent of its sampled provisions to Rules of Origin, against 24 percent for RCEP and 39 percent for AANZFTA. Tariff Commitments tells a parallel story at 31 percent for AHKFTA against 6 percent for RCEP. Together these proportions indicate that AHKFTA's provision space is compressed into the chapters that goods-trade agreements include, and within that compressed space, origin regulation absorbs a far larger share of the legal text than in the broader integration agreements.
- **RCEP and AHKFTA share a Regional Value Content threshold but split on the transformation rule.** Both apply a 40 percent RVC test, which makes them look interchangeable at a headline level. The substantive divergence sits one step below: RCEP applies a Change in Tariff Heading (CTH) test at the 4-digit HS level, while AHKFTA applies a Change in Chapter (CC) test at the 2-digit HS level. CC is the stricter test, requiring transformation across a broader category of goods, which means the same product can satisfy RCEP's origin rule and fail AHKFTA's even when the value-content figure is identical.
- **Customs Procedures is the only category genuinely converging across all three agreements.** Its normalised entropy ratio reaches 1.00, the highest in the matrix, consistent with the WTO Trade Facilitation Agreement providing a shared procedural baseline that all three parties follow on documentation, advance rulings, and risk-based release. Dispute Settlement (entropy 0.47) and Intellectual Property (0.37) sit at the fragmented end, but the fragmentation reflects scope difference rather than drafting conflict: AHKFTA simply does not include those chapters, so an entropy metric that cannot distinguish "absent" from "disagree" labels them as fragmented even though RCEP and AANZFTA do not write them in incompatible ways.
- **AANZFTA leads on dispute settlement intensity and is the only agreement of the three that maintains an ISDS mechanism.** Its 22 sampled provisions in Dispute Settlement, against 6 for RCEP and 0 for AHKFTA, reflect a fully independent adjudication and arbitration architecture, separate from the WTO DSU.

The most policy-actionable result is the CC-versus-CTH difference. Two agreements can read as nearly identical at the threshold level because both demand 40 percent regional value content, yet still impose meaningfully different compliance burdens through the structure of their tariff-classification tests. A manufacturer who has structured a supply chain to clear RCEP's CTH rule cannot assume the same supply chain will clear AHKFTA's CC rule, and on paper the same goods can qualify for preferential rates under one agreement and fail under the other. That is exactly the kind of subtle, document-buried distinction that analysts and customs officers most often need surfaced quickly, and it is precisely what a structured comparison layer over LLM-classified provisions is well positioned to flag for closer manual review. Beyond the headline finding, the broader implication is that procedural harmonisation across the region appears to be advancing on a shared template, while substantive integration on services, investment, dispute settlement, and IP remains uneven and largely shaped by which chapters each agreement chose to include in the first place.

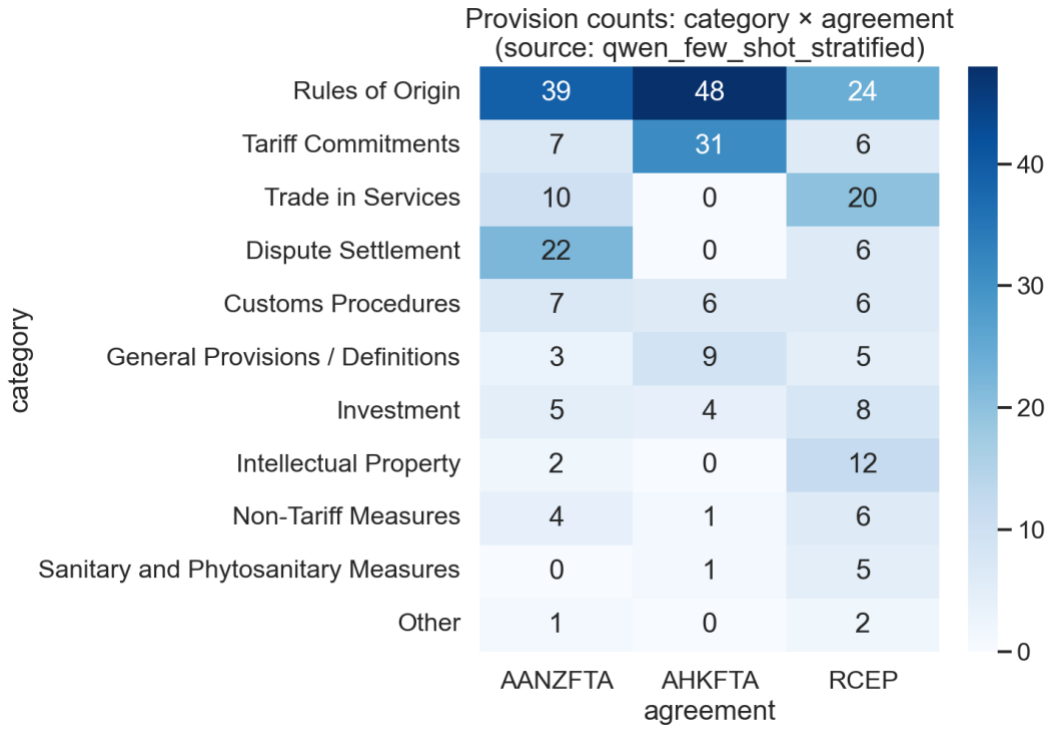

Figure 3. Category distribution by agreement on the 100-per-agreement stratified cohort.

Conclusion

Across the three research questions, the project produced defensible answers without overclaiming. RQ1 confirmed that open-weight LLMs can classify FTA provisions, but only at triage-grade accuracy (32 to 48 percent on the gold set), with prompt strategy choices that have to be calibrated per model rather than universally. RQ2 surfaced concrete, observable design differences between the three agreements, of which the CC versus CTH transformation rule sitting on top of an identical 40 percent RVC threshold is the most policy-actionable. RQ3 showed a clear pattern of selective convergence: Customs Procedures is the only category genuinely in sync across all three agreements, while Dispute Settlement and Intellectual Property fragment mainly because AHKFTA does not include those chapters at all. Read together, these results support a conditional yes on feasibility. Using LLMs to compare Free Trade Agreements, and by extension other cross-country legal instruments with similarly structured chapters and shared terminology, is feasible today as a first-pass triage layer that helps analysts prioritise where to look. It is not yet feasible as a standalone classifier for compliance or legal interpretation, and any deployment that frames it that way overstates what the current evidence supports.

Recommendation

The strongest next step is not a larger model but a stronger evaluation foundation. Expanding the hand-labelled gold set well beyond fifty provisions, introducing a second human annotator to quantify human-level disagreement, and adopting a hierarchical taxonomy that separates definitional, procedural, and substantive provisions would all raise the ceiling on what future runs can defensibly claim. On the data side, integrating tariff schedules, product-specific annexes, and adjacent legal instruments (such as the ASEAN-Hong Kong Investment Area Agreement) into the corpus would close the most visible scope gaps and let convergence findings rest on a fuller picture of each treaty system. Beyond a single regional bloc, broadening the comparison to include CPTPP, AFTA, and major bilateral agreements would test whether the patterns observed here generalise. Methodologically, future iterations should test fine-tuning a smaller open-weight model on the FTA-specific taxonomy to reduce reliance on large-model APIs, build per-provision meta-classifiers that exploit complementary error structures across runs, and pilot the workflow with a real trade-policy analyst to verify that the time savings survive contact with operational use. The shorter version: invest in better labels, broader corpus, and stronger evaluation discipline before chasing higher headline metrics, and the same pipeline can credibly extend from FTAs to other cross-country legal corpora that share their structural shape.